

Epitope	No. of substitutions	Top 40 high-effect HA1 substitutions
A	5	K131T, S137F*, N145S, S144R, I140R
B	13	K189R, N158K, G186S, N190T*, S193F, V186G, N159Y, H156Q, F159Y, P194L, S156H, L194P, Q197R
C	5	K276N, E50K, E276K, D53N, S54R
D	5	N225D*, R222K, R201I, Q173R, E172G
E	4	Q75R, E62K, T65K, K62E
Other epitope	1	I202V
Unannotated	7	H183L, M168V, V223I, S228T, Y233H, G49D, K310R